22. More Stochastic Approximation

March 6, 2026 9:29 AM

Main Goal is to find θ^* to minimize some Loss Function $Q(\theta, v)$, with noise v.

RM	KW
requires the gradient $y(\theta)$	Estimates the gradient $\hat{y}(\theta)$

d = dimension

used $2 \cdot d$ Loss function evaluations

Other Gradient-Free Methods that <u>don't</u> require $2d$ computations of the loss include

- Random Perturbations
- Simultaneous Perturbations

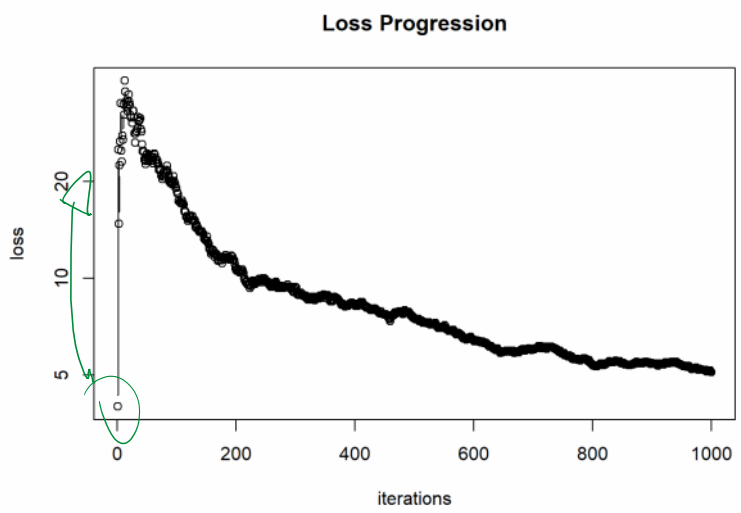
} pick a random direction in $\mathbb{R}^d$ & compute the gradient in that direction

a "Directional" derivative

→ Pro: only 2 evaluations of Q

→ Con: the "random" direction may be very poor.

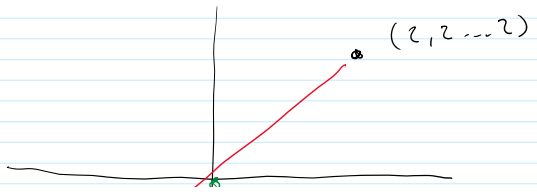
Example: optimize $\sum_{i=1}^{6} \theta_i^2$ in $\mathbb{R}^6$



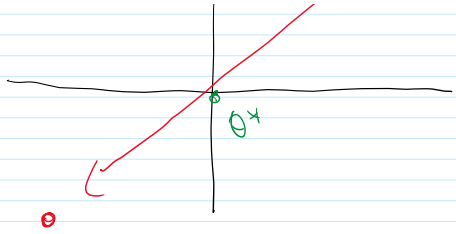
- Started at $\theta_0 = (2, \dots, 2)$
- RM says $\nabla L(\theta) = 2 \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_6 \end{pmatrix}$

  $\nabla L(\theta_0) = \begin{pmatrix} 4 \\ \vdots \\ 4 \end{pmatrix}$
- $a_k = 1/k \cdot \text{const}$

$\therefore \theta_1 = \theta_0 - \frac{1}{1} \cdot \text{const} \begin{pmatrix} 4 \\ \vdots \\ 4 \end{pmatrix}$

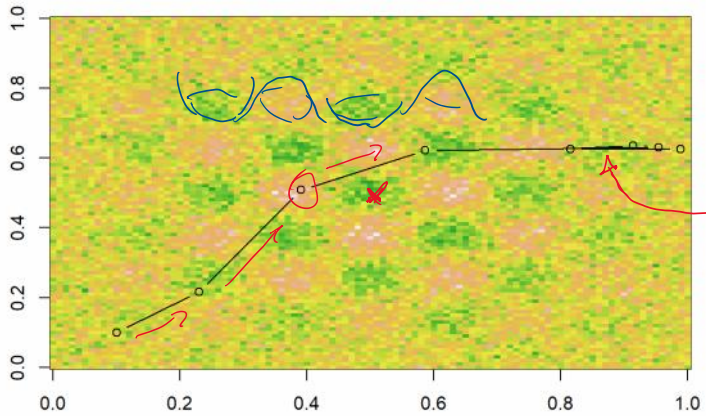


Point: Unlike, say, LRS RM or KW could "miss" the optimum and increase



RM or KW could "miss" the optimum and increase the loss.

over shoot / Miss the optimum θ^*

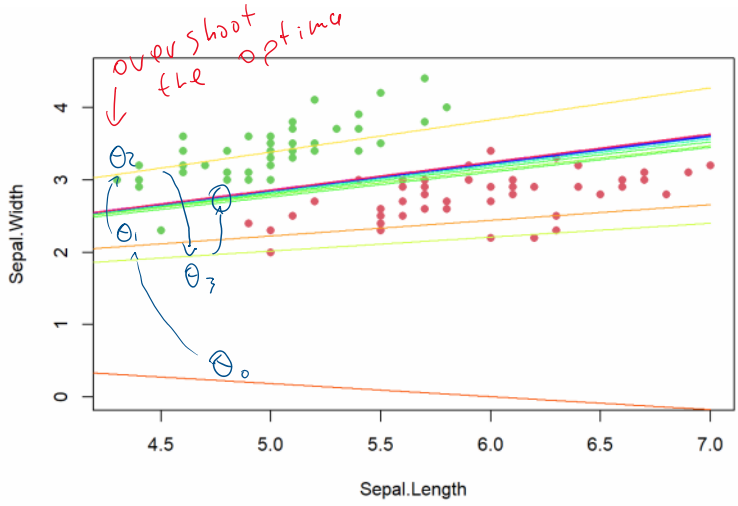


Stuck in local min

Point: Methods like RM, KW get stuck in local minima

However, Many problems in Stats, ML, Data Science are convex opt.

SVM Fit with RM



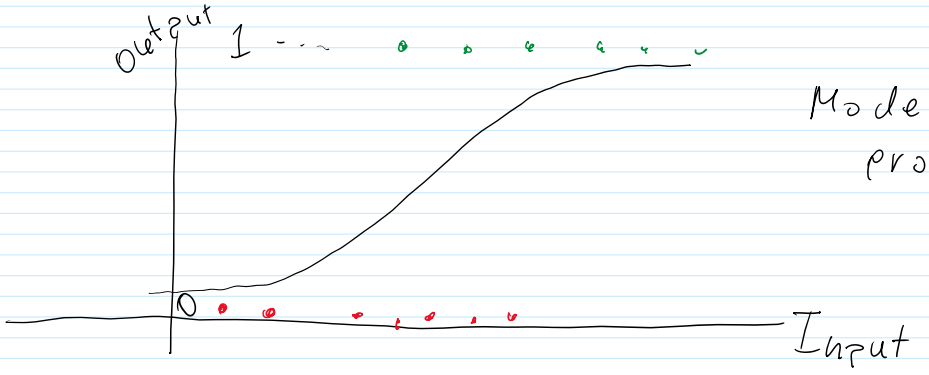
One Sense

→ opt. in $\mathbb{R}^3$

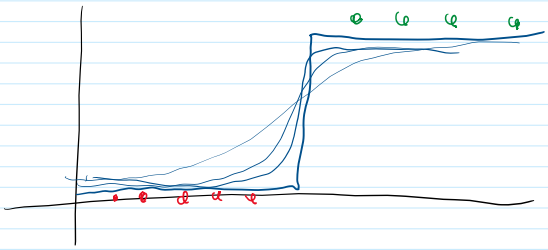
i.e. Find $\theta^* \in \mathbb{R}^3$ to min some Loss

→ each vector θ_k corresponds to a line (Hyperplane)

We can also try to Fit Logistic Regression



Model the prob(Green | Input)

SVM just trying to seperate points with a Hyperplane

> SVM just tries to seperate points with a Hyperplane